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Lab 8: Calculator

Today we will replace last week's circuit such that it functions as a calculator.

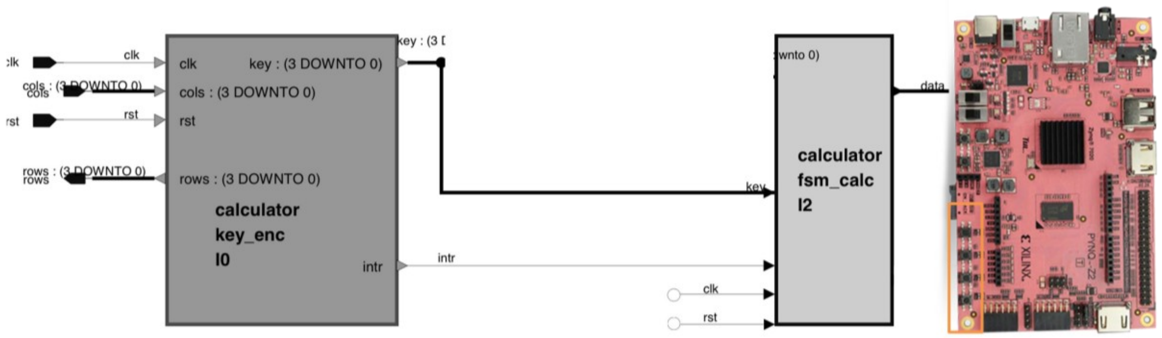
Discussion – Lab 7

1. Any issues?

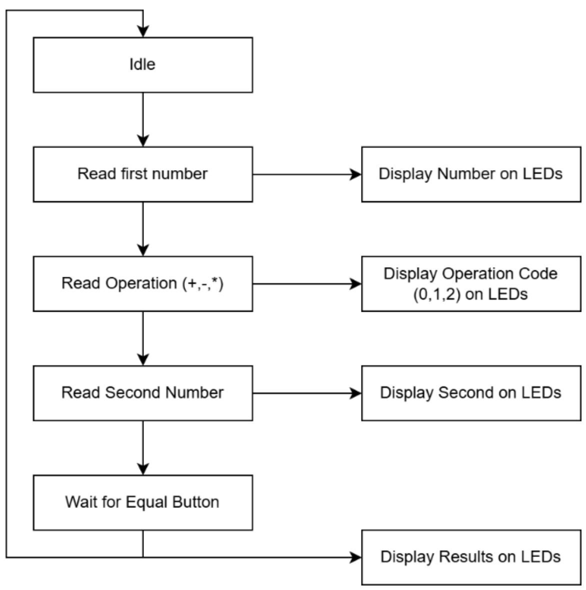
Continuation Lab 7

1. If you haven't finished Lab 7, work on Lab 7

Task – Calculator



1. Consider the two schematic, the Task is to build a simple Calculator, that reads the keypad inputs, calculates and displays intermediate values and results with the LEDS (in binary)
2. Draw out the state machine diagram for the FSMs for a simple calculator that adds two input numbers
3. Implement the FSMs in VHDL
4. Test and verify it individually in simulation
5. Include the Keypad driver from previous lab sessions and make the full circuit in a block design
6. Synthesize, implement, generate bitstream, and program your FPGA
7. Verify the functionality of the circuit
8. If it doesn't work, try debugging with post-synthesis and post- implementation simulation
9. Repeat the steps above for adding subtraction and multiplication to the system



Credit to original author Frederik Falk Nyboe, ffn@mmmi.sdu.dk